

RRC-Raptor-V4H PCle

- User's Manual: Software

V0.0.1

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i. VERSION LIST

Description	Version	Date	Author
RTX_PCIE	V0.0.1	2025/01/09	Jerry

ii. HASHTAG

#V4H

#PCle

#BSP

1. Overview

1.1. Overview

This manual explains the driver module (this module) that controls the PCIe Controller on R-Car.

1.2. Function

This module controls the PCIe Controller on R-Car V4H, acting as a PCIe Host Controller. It allows PCIe Endpoint devices to be used with R-Car V4H . The following functionality is supported:

- Configuration read and writes.
- Host initiated IO (legacy) reads and writes.
- Host initiated memory reads and writes.
- Slave initiated memory reads and writes (bus mastering).
- MSI interrupts.

❖ For R-Car V4H

This module supports dual lane and two channels:

- Support two dual lane PCIE0 & PCIE1 and one quad lane

1.3. Connected Port

Switch PCIe mode through SW3 as follows:

SW3	3:PCIE_MUX	2:PCIE_SEL	Connect
1-2	Don't Care	H(P3.3V)	A to C M.2 Key M(HSSSTP)
Center (DEFAULT)	Don't Care	L	A to B PCIe SW Connect
2-3	L	L	A to B PCIe SW Connect
	H	H	A to C M.2 Key M(HSSSTP)

Use the corresponding port based on the selected mode.

Connect	PCIE	LANE	PORT	Enable / Disable
A to C M.2 Key M(HSSSTP)	PCIE0	4 lanes	CN14 (Key M)	Enable
	PCIE1	Disable	CN15 (Key M) J1 (Key E) CN13 (OcuLink) UAB1 (Type C) UAB2 (Type A) UAB3 (Type A) UAB4 (Type A)	Disable Disable Disable Disable Disable Disable Disable
A to B PCIe SW Connect	PCIE0	2 lanes	CN14 (Key M)	Enable
	PCIE1	2 lanes	CN15 (Key M) J1 (Key E) CN13 (OcuLink) UAB1 (Type C) UAB2 (Type A) UAB3 (Type A) UAB4 (Type A)	Enable Enable Enable Enable Enable Enable Enable

2. Directory Configuration

The directory configuration is shown below.

```
—— drivers/pci/controller/dwc/  ——  pcie-rcar-gen4-host.c  : PCIe v4.0 Host Controller source file

—— drivers/pci/controller/dwc/  ——  pcie-rcar-gen4-ep.c   : PCIe v4.0 Endpoint source file
```

Figure 1. Directory Configuration

3. Kernel configuration

To enable the Root Complex mode of this module, make the following setting with Kernel Configuration. This module is selected only built-in module.

```
Device Drivers --->
  [*] PCI support
    *- Message Signaled Interrupts (MSI and MSI-X)
      PCI controller drivers --->
        DesignWare PCI Core Support --->
          <*> Renesas R-Car Gen4 PCIe Host controller

Device Drivers --->
  Misc devices --->
    [*] PCI Endpoint Test driver
```

Figure 2 Kernel configuration

4. Switch For Max link speed (Gen)

Modify the max-link-speed of pciec0/pciec1 in r8a779g0.dtsi to set the maximum link speed (Gen).

```
pciec0: pcie@65d0000 {
    compatible = "renesas,r8a779g0-pcie",
        "renesas,rcar-gen4-pcie";
    reg = <0 0xe65d0000 0 0x3000>, <0 0xe65d3000 0 0x2000>,
        <0 0xe65d5000 0 0x1200>, <0 0xe65d6200 0 0x0e00>,
        <0 0xe65d7000 0 0x1000>, <0 0xfe000000 0 0x4000000>;
    reg-names = "dbi", "atu", "dma", "app", "phy", "config";
    #address-cells = <3>;
    #size-cells = <2>;
    bus-range = <0x00 0xff>;
    device_type = "pci";
    ranges = <0x82000000 0 0x30000000 0 0x30000000 0 0x10000000>;
    dma-ranges = <0x42000000 0 0x40000000 0 0x40000000 0 0x80000000>;
    interrupts = <GIC_SPI 449 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 450 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 451 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 452 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 453 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 454 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 455 IRQ_TYPE_LEVEL_HIGH>;
    interrupt-names = "others", "dma", "correctable",
        "fatal", "nonfatal", "lp", "vndmsg";
    #interrupt-cells = <1>;
    interrupt-map-mask = <0 0 0 7>;
    interrupt-map = <0 0 0 1 &gic GIC_SPI 449 IRQ_TYPE_LEVEL_HIGH
        0 0 0 2 &gic GIC_SPI 449 IRQ_TYPE_LEVEL_HIGH
        0 0 0 3 &gic GIC_SPI 449 IRQ_TYPE_LEVEL_HIGH
        0 0 0 4 &gic GIC_SPI 449 IRQ_TYPE_LEVEL_HIGH>;
    clocks = <&cpg CPG_MOD 624>;
    power-domains = <&sysc R8A779G0_PD_ALWAYS_ON>;
    resets = <&cpg 624>;
    num-lanes = <2>;
    snps,enable-sdm-check;
    max-link-speed = <4>;
    status = "disabled";
};
```

```
pciec1: pcie@65d8000 {
    compatible = "renesas,r8a779g0-pcie",
        "renesas,rcar-gen4-pcie";
    reg = <0 0xe65d8000 0 0x3000>, <0 0xe65db000 0 0x2000>,
        <0 0xe65dd000 0 0x1200>, <0 0xe65de200 0 0x0e00>,
        <0 0xe65df000 0 0x1000>, <0 0xee900000 0 0x4000000>;
    reg-names = "dbi", "atu", "dma", "app", "phy", "config";
    #address-cells = <3>;
    #size-cells = <2>;
    bus-range = <0x00 0xff>;
    device_type = "pci";
    ranges = <0x82000000 0 0xc0000000 0 0xc0000000 0 0x10000000>;
    dma-ranges = <0x42000000 0 0x40000000 0 0x40000000 0 0x80000000>;
    interrupts = <GIC_SPI 456 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 457 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 458 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 459 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 460 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 461 IRQ_TYPE_LEVEL_HIGH>,
        <GIC_SPI 462 IRQ_TYPE_LEVEL_HIGH>;
    interrupt-names = "others", "dma", "correctable",
        "fatal", "nonfatal", "lp", "vndmsg";
    #interrupt-cells = <1>;
    interrupt-map-mask = <0 0 0 7>;
    interrupt-map = <0 0 0 1 &gic GIC_SPI 456 IRQ_TYPE_LEVEL_HIGH
        0 0 0 2 &gic GIC_SPI 456 IRQ_TYPE_LEVEL_HIGH
        0 0 0 3 &gic GIC_SPI 456 IRQ_TYPE_LEVEL_HIGH
        0 0 0 4 &gic GIC_SPI 456 IRQ_TYPE_LEVEL_HIGH>;
    clocks = <&cpg CPG_MOD 625>;
    power-domains = <&sysc R8A779G0_PD_ALWAYS_ON>;
    resets = <&cpg 625>;
    num-lanes = <2>;
    snps,enable-sdm-check;
    max-link-speed = <4>;
    status = "disabled";
};
```

The configuration values and their corresponding speeds are as follows:

Max-link-speed	Gen	Speed
1	1	2.5 GT/s
2	2	5.0 GT/s
3	3	8.0 GT/s
4	4	16.0 GT/s

5. Connect

Provide multiple PCIe interfaces so that users can add their own PCIe devices.

● A to C Mode :

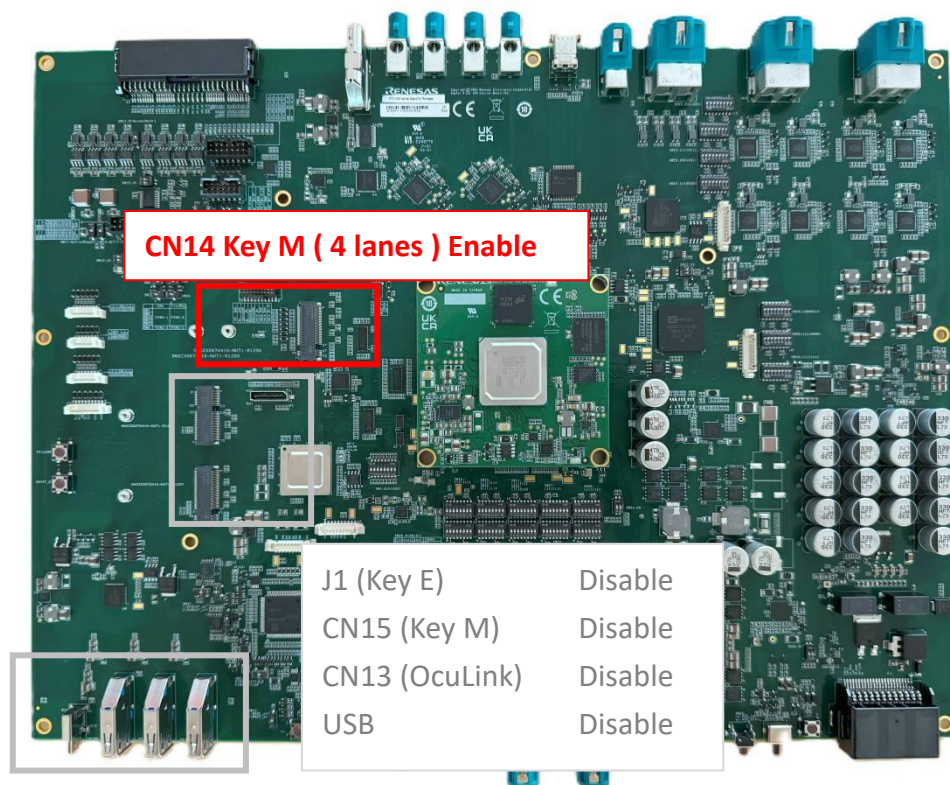


Figure 3 Connect

● **A to B Mode :**

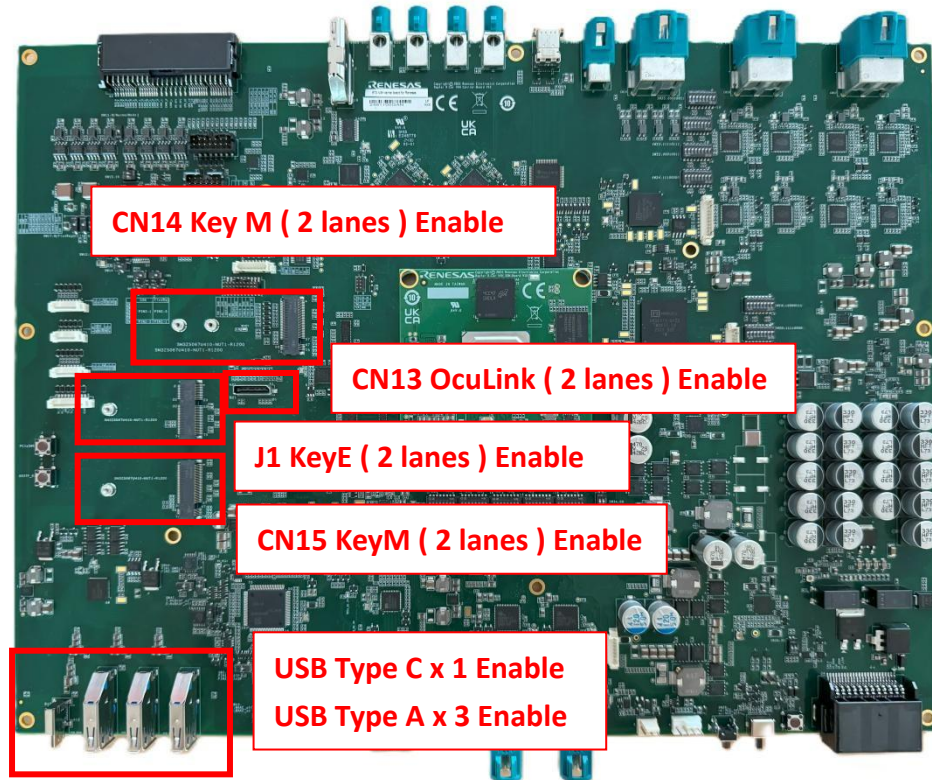


Figure 4 Connect

6. Verify

A to B Mode :

1. Insert a 4-lane NVMe M.2 SSD into CN14.
2. Use lspci to check the SSD device and link status information.

```
root@v4x:~# lspci
00:00.0 PCI bridge: Renesas Technology Corp. Device 0030
01:00.0 Non-Volatile memory controller: Device 1d79:2269 (rev 03)

root@v4x:~# lspci -vvv -s 01:00.0
01:00.0 Non-Volatile memory controller: Device 1d79:2269 (rev 03) (prog-if 02 [NVM Express])
Subsystem: Device 1d79:2269
Control: I/O- Mem+ BusMaster+ SpecCycle- MemWINV- VGASnoop- ParErr- Stepping- SERR- FastB2B- DisINTx+
Status: Cap+ 66MHz- UDF- FastB2B- ParErr- DEVSEL=fast >TAbort- <TAbort- <MAbort- >SERR- <PERR- INTx-
Latency: 0
Interrupt: pin A routed to IRQ 454
NUMA node: 0
Region 0: Memory at 30000000 (64-bit, non-prefetchable) [size=16K]
Capabilities: [40] Power Management version 3
Flags: PMEClk- DSI- D1- D2- AuxCurrent=0mA PME(D0-,D1-,D2-,D3hot-,D3cold-)
Status: D0 NoSoftRst+ PME-Enable- DSel=0 DScale=0 PME-
Capabilities: [50] MSI: Enable+ Count=8/16 Maskable+ 64bit+
Address: 000000004a41e000 Data: 0008
Masking: 0000ffe0 Pending: 00000000
Capabilities: [70] Express (v2) Endpoint, MSI 00
DevCap: MaxPayload 256 bytes, PhantFunc 0, Latency L0s unlimited, L1 unlimited
ExtTag- AttnBtn- AttnInd- PwrInd- RBE+ FLReset+ SlotPowerLimit 0.000W
DevCtl: CorrErr- NonFatalErr- FatalErr- UnsupReq-
RlxdOrd+ ExtTag- PhantFunc- AuxPwr- NoSnoop- FLReset-
MaxPayload 256 bytes, MaxReadReq 256 bytes
DevSta: CorrErr- NonFatalErr- FatalErr- UnsupReq- AuxPwr- TransPend-
LnkCap: Port #0, Speed 16GT/s, Width x4, ASPM L1, Exit Latency L1 <64us
ClockPM+ Surprise- LLActRep- BwNot- ASPMOptComp+
LnkCtl: ASPM Disabled; RCB 64 bytes Disabled- CommClk-
ExtSynch- ClockPM- AutWidDis- BWInt- AutBWInt-
LnkSta: Speed 16GT/s (ok), Width x4 (ok)
TrErr- Train- SlotClk+ DLActive- BWMgmt- ABWMgmt-
```

3. Mount the SSD to /mnt.

```
root@v4x:~# ls /dev/nvme*
/dev/nvme0 /dev/nvme0n1
root@v4x:~# mount /dev/nvme0n1 /mnt/
[ 1414.351039] EXT4-fs (nvme0n1): mounted filesystem with ordered data mode. Opts: (null)
root@v4x:~# mount | grep nvme0n1
/dev/nvme0n1 on /mnt type ext4 (rw,relatime)
```

4. Perform read/write tests on it

```
root@v4x:~# dd if=/dev/random of=./test_file bs=1024 count=100
100+0 records in
100+0 records out
102400 bytes (102 kB, 100 KiB) copied, 0.00436699 s, 23.4 MB/s
root@v4x:~# md5sum /mnt/test_file
d989fa26f278f633818588207607f4ae /mnt/test_file
root@v4x:~# cp /mnt/test_file ./
root@v4x:~# md5sum ./test_file
d989fa26f278f633818588207607f4ae ./test_file
```

5. Unmount the SSD.

```
root@v4x:~# umount /dev/nvme0n1
root@v4x:~# mount | grep nvme0n1
```

6. Confirm that the link status is **4 lanes** and that read/write operations on the SSD are functioning properly.

A to C Mode :

1. Connect the verification device.
 - I. Insert NVMe M.2 SSDs into CN14 and CN15.
 - II. Connect an Oculink adapter to CN13, then attach an NVMe M.2 SSD.
 - III. Insert a WiFi/BT device with a Key E interface into J1.
 - IV. Connect four USB storage devices.

2. 用 lspci / lsusb 查看以上裝置是否有連接

```
root@v4x:~# lspci
0000:00:00.0 PCI bridge: Renesas Technology Corp. Device 0030
0000:01:00.0 Non-Volatile memory controller: Device 1d79:2269 (rev 03)
0001:00:00.0 PCI bridge: Renesas Technology Corp. Device 0030
0001:01:00.0 PCI bridge: Pericom Semiconductor Device c016 (rev 07)
0001:02:01.0 PCI bridge: Pericom Semiconductor Device c016 (rev 06)
0001:02:02.0 PCI bridge: Pericom Semiconductor Device c016 (rev 06)
0001:02:03.0 PCI bridge: Pericom Semiconductor Device c016 (rev 06)
0001:02:04.0 PCI bridge: Pericom Semiconductor Device c016 (rev 06)
0001:02:05.0 PCI bridge: Pericom Semiconductor Device c016 (rev 06)
0001:02:06.0 PCI bridge: Pericom Semiconductor Device c016 (rev 06)
0001:02:07.0 PCI bridge: Pericom Semiconductor Device c016 (rev 06)
0001:03:00.0 Non-Volatile memory controller: Samsung Electronics Co Ltd NVMe SSD Controller SM981/PM981/PM983
0001:04:00.0 Ethernet controller: Marvell Technology Group Ltd. Device 2b43 (rev 03)
0001:04:00.1 Ethernet controller: Marvell Technology Group Ltd. Device 2b44 (rev 03)
0001:04:00.2 Bluetooth: Marvell Technology Group Ltd. Device 2b45 (rev 03)
0001:05:00.0 Non-Volatile memory controller: Micron Technology Inc Device 6001 (rev 03)
0001:06:00.0 USB controller: Renesas Technology Corp. uPD720201 USB 3.0 Host Controller (rev 03)
root@v4x:~# m/
```

CN14 :

0000:01:00.0 **Non-Volatile memory** controller: Device 1d79:2269 (rev 03)

CN15 :

0001:05:00.0 **Non-Volatile memory** controller: Micron Technology Inc Device 6001 (rev 03)

CN13 :

0001:05:00.0 **Non-Volatile memory** controller: Micron Technology Inc Device 6001 (rev 03)

J1 :

0001:04:00.0 **Ethernet** controller: Marvell Technology Group Ltd. Device 2b43 (rev 03)

0001:04:00.1 **Ethernet** controller: Marvell Technology Group Ltd. Device 2b44 (rev 03)

0001:04:00.2 **Bluetooth**: Marvell Technology Group Ltd. Device 2b45 (rev 03)

```
root@v4x:~# lsusb
Bus 002 Device 002: ID 13fe:5500 Kingston Technology Company Inc. Silicon-Power32G
Bus 002 Device 004: ID 8564:1000 Transcend Information, Inc. JetFlash
Bus 002 Device 003: ID 125f:db8a A-DATA Technology Co., Ltd. ADATA USB Flash Drive
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 002: ID 8564:1000 Transcend Information, Inc. JetFlash
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

Four USB storage devices :

Bus 002 Device 002: ID 13fe:5500 Kingston Technology Company Inc. Silicon-Power32G

Bus 002 Device 004: ID 8564:1000 Transcend Information, Inc. JetFlash

Bus 002 Device 003: ID 125f:db8a A-DATA Technology Co., Ltd. ADATA USB Flash Drive

Bus 001 Device 002: ID 8564:1000 Transcend Information, Inc. JetFlash

3. The storage devices can be seen under /dev

```
root@v4x:~# ls -l /dev/nvme*
crw----- 1 root root 508, 0 Jan  9 02:03 /dev/nvme0
brw-rw---- 1 root disk 259, 0 Jan  9 02:03 /dev/nvme0n1
crw----- 1 root root 508, 1 Jan  9 02:03 /dev/nvme1
brw-rw---- 1 root disk 259, 1 Jan  9 02:03 /dev/nvme1n1
crw----- 1 root root 508, 2 Jan  9 02:03 /dev/nvme2
brw-rw---- 1 root disk 259, 2 Jan  9 02:03 /dev/nvme2n1
root@v4x:~# ls -l /dev/sd*
brw-rw---- 1 root disk 8,  0 Jan  9 02:03 /dev/sda
brw-rw---- 1 root disk 8,  1 Jan  9 02:03 /dev/sda1
brw-rw---- 1 root disk 8, 16 Jan  9 02:03 /dev/sdb
brw-rw---- 1 root disk 8, 17 Jan  9 02:03 /dev/sdb1
brw-rw---- 1 root disk 8, 32 Jan  9 02:03 /dev/sdc
brw-rw---- 1 root disk 8, 33 Jan  9 02:03 /dev/sdc1
brw-rw---- 1 root disk 8, 48 Jan  9 02:03 /dev/sdd
brw-rw---- 1 root disk 8, 49 Jan  9 02:03 /dev/sdd1
```

4. Use the previous method to mount the storage and perform read/write tests.

7. APPENDIX